ASE A3 Simple Sheet — Manual Drive Train and Axles

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A condensed, four-page middle ground between the full study guide and the one-page cram sheet: enough explanation to reason through a Tech A/B or "most likely" question, without the full guide's setup-procedure depth. Official weights: Clutch 16% · Transmission 17% · Transaxle 17% · Shafts/joints 13% · Drive axle 17% · 4WD/AWD 20%. 50 questions, 40 scored, 60 minutes — unscored items aren't identified, so treat every question as scored.

A. Clutch — 16%

Power path (engaged): flywheel → clamped disc → disc splines → input shaft. **Release path:** pedal/linkage or hydraulic pressure → release bearing → pressure-plate fingers → clamp force removed.

The two failure directions are opposites, and exam questions hinge on telling them apart:

- **Slip** — the clutch will not *hold* torque. Shows up as RPM flare under load. Causes: worn/glazed friction material, contamination, weak clamp load, or too little free play holding the release bearing against the fingers.
- **Drag** — the clutch will not *release*. Shows up as hard engagement or clash into first/reverse while stopped. Causes: incomplete release travel (too much free play, air in hydraulics, worn linkage), warped disc, or a bound pilot bearing.

Pulsation at the pedal points to a *rotating* component (flywheel runout, warped pressure plate/disc) rather than the hydraulic system. Always check the rear main seal and other contamination sources before a new clutch goes in, and never air-blow clutch dust.

Chatter is a separate cause map from slip/drag — don't reach for free play here. The classic chatter cause is contamination (oil/grease/coolant) on the friction disc; uneven pressure-plate fastener torque and a low clutch master-cylinder fluid level can also cause it.

Release-bearing noise timing depends on the design. A traditional pedal-actuated release bearing only contacts the pressure-plate fingers when the pedal is pressed, so it's noisiest *disengaged*. A continuous-contact/self-adjusting (SAC) release bearing, common on GM vehicles, rotates with the engine essentially all the time, so it's noisiest *engaged* — identify the design before picking an answer. A 15-20 mph full-throttle road test in high gear checks both slippage and spin-down, and air in a hydraulic clutch circuit is a drag cause that calls for a bleed after repair.

B. Transmission — 17%

The constant-mesh synchronizer sequence is the single most important sentence in this category:

Fork moves the sleeve → sleeve carries the keys/struts against the blocker ring → the ring's friction cone matches gear speed to shaft speed → once speeds match, the sleeve slides past the ring and locks onto the gear's dog teeth — not the cone. The cone only synchronizes speed; the dog teeth carry the actual torque.

Reading clash symptoms: if only *one* gear clashes, suspect that gear's synchronizer specifically. If clash happens shifting into *every* gear, look at something shared by all of them first — clutch release, linkage adjustment, or fluid — before condemning every synchro at once. Pop-out of gear points to engagement depth, worn detent, or shaft endplay rather than the synchronizer itself.

Don't overlook the less glamorous officially-tested items: gaskets, seals, and sealing surfaces at every mating joint; case/extension-housing bores, dowels, and **vents** (a blocked vent pressurizes the case and can push fluid past a seal that's perfectly fine); and mount inspection, since a failed mount can mimic an internal noise or vibration complaint.

C. Transaxle — 17%

Everything from Section B applies — same synchronizer logic, same clash/pop-out reasoning, same gasket/seal/vent/mount coverage. What's added is that the final drive and differential live in the same case as the transmission, so the differential logic from Section E (below) is in scope here too — and specifically, the transaxle's final drive is tested on the **same preload/backlash/contact-pattern setup logic as Section E's rear axle**, just in a different housing, not a separate topic. Shaft/component naming doesn't automatically carry over from a rear-wheel-drive transmission — confirm the specific unit's input/output arrangement rather than assuming. Fluid level, the lubrication system, and sensor/actuator/control-module circuits are officially tested here too, same as Section B.

D. Drive/Half-Shafts and U/CV Joints — 13%

Symptom map: clicking/popping in turns → outer CV joint (it's the fixed one, nearest the wheel). Shudder on acceleration → inner/plunging CV joint (nearest the transmission or transfer case). Clunk on accel/decel → driveline lash, splines, or a slip joint. Speed-related vibration that isn't obviously a CV joint → balance, runout, working angle, or phasing.

Why a U-joint vibrates at all: a standard cross-type (Cardan) joint's output speed varies *twice per revolution* because the joint's effective radius changes as it rotates through an angle — bigger angle, bigger variation. This is called 2nd-order vibration (a multiple of shaft rotation speed, as opposed to 1st-order, which tracks shaft speed directly). It's inherent to the design, not automatically a fault.

Canceling it on a two-joint shaft takes two separate things, and missing either one is the classic way to get this wrong:

1. **Phasing** — the yokes at each end are oriented so the two joints' speed-ups and slow-downs happen out of step with each other, commonly described as phased about 180° apart.
2. **Equal working angles** — the operating angle at each joint has to match the other (commonly cited within about half a degree), or cancellation is incomplete even with correct phasing.

The **working angle** itself is the angle *between the two members a joint connects* — for example, the transmission/transfer-case output shaft versus the driveshaft, or the driveshaft versus the pinion yoke — not just how much the shaft appears to slope off level ground. It's measured with an angle finder on each member's reference surface, and the working angle at a joint is the *difference* between those two readings. A typical acceptable range is roughly 0.5°–3°; a 0° angle is also wrong, because the joint needs a little rocking motion to keep distributing bearing load instead of sitting static. Vertical and horizontal misalignment together (a compound angle) don't simply add — a shaft can pass a vertical-only check and still be out of tolerance once horizontal offset is included.

CV joints don't have this problem at all — they're built to hold output speed constant through their operating angle, which is why they sit at the wheel end instead of a plain cross-type joint.

E. Drive Axle Diagnosis and Repair — 17%

Power flow: pinion → ring gear → differential case → spider/pinion gears → side gears → axle shafts. Straight ahead, the side gears turn together; in a turn, the spider gears roll on their shaft so the outer wheel can turn faster than the inner one. On an **open** differential, a broken axle shaft can cause a no-drive condition on **both** wheels — the broken side spins free and the intact side never gets reaction torque.

Tooth geometry vocabulary: heel = outer (large) end of the tooth, toe = inner (small) end; face = above the pitch line, flank = below it.

Hunting vs. non-hunting: if the ring/pinion tooth counts share no common factor (the ratio doesn't reduce — e.g. 41/11), it's a **hunting** set — a given pinion tooth eventually meets every ring-gear tooth over enough revolutions, and no factory timing marks are needed. If the ratio reduces to a whole number (e.g. 39/13 = 3), it's **non-hunting** — a repeating, limited subset of tooth contacts, so factory timing marks matter at reassembly to preserve the original break-in pattern.

Reading a contact pattern is two separate adjustments controlling two separate axes — this is the piece worth memorizing cold:

- **Pinion depth** moves the pattern toward the **face** (shallow mesh, pattern rides high) or the **flank** (deep mesh, pattern rides low/toward the root). High on the face → pinion is too far from the ring gear → move it **in**. Low on the flank → pinion is too close → move it **out**.
- **Backlash**, adjusted at the side bearings (which moves the ring gear itself), moves the pattern toward the **toe** (too much backlash) or the **heel** (too little backlash). Toward toe → move the ring gear **in** to reduce backlash. Toward heel → move it **out** to increase backlash.

Always read the pattern on the drive side, expect the coast side to move consistently with the correction, and finish against the manufacturer's own illustration — this is the general mnemonic, not a substitute for the axle-specific chart.

Setup basics: backlash is tooth clearance (dial indicator at the ring gear, check several positions). Preload is bearing load, and it's confirmed by *rotating torque*, not the wrench's tightening torque. A crush sleeve collapses once to set that preload — if you overshoot the spec, the fix is a new sleeve, not backing off the nut.

Axle support types: semi-floating shafts carry both torque and vehicle weight; full-floating shafts carry mainly torque, with the hub bearings carrying weight into the housing.

Traction-device map, broadest to most specific: open differential (possibly paired with brake-based traction control) → clutch-type limited slip/Posi (clutch packs resist side-speed difference, need the specified friction-modified fluid) → helical/Torsen limited slip (gear-geometry torque biasing, no clutch pack — don't apply clutch-LSD logic to it) → GM/Eaton-style automatic mechanical locker (a low-speed governor/cam mechanism engages full lock) → electric or air locker (a commanded actuator produces full lock on demand). “Limited slip” or “locker” alone isn't enough identification — find the actual mechanism before choosing a test.

Noise map: whine/howl that changes between drive and coast = ring/pinion contact or setup. Rough growl/rumble = a bearing. Growl that changes with cornering load = wheel bearing. Clunk on torque reversal = backlash/lash or worn splines/joints. Chatter on tight turns = limited-slip fluid or clutch condition.

F. Four-Wheel Drive / All-Wheel Drive — 20% (the largest single category)

Start by naming the architecture, because the right diagnostic path depends on it:

- **Part-time 4WD** — front and rear are positively coupled with no center differential. On dry pavement in a turn, the axles want different average speeds; since the system can't provide that, torque winds up in the driveline. Bind, hop, or hard disengagement in this situation can be normal for the architecture, not automatically a fault.
- **Full-time 4WD/AWD** — a center differential or controlled coupling (geared, viscous, hydraulic, electromagnetic, or multi-plate clutch) manages the front/rear speed difference continuously.
- **On-demand AWD** — one axle is normally the primary driver; a controlled coupling adds the second axle when needed, and engagement can be commanded preemptively from throttle position, steering angle, or a modeled demand — not only in response to actual wheelspin.

Two forks, two jobs — the point Logan's final tested directly:

- **Range fork** selects high range (near direct) or low range (torque multiplication/reduction). Purely a gearing decision.
- **Mode fork** selects 2WD or 4WD — whether the secondary output is coupled at all. Purely a coupling decision.

They're driven by separate actuators. 2HI and 4HI share the same range and differ only in whether the mode fork has coupled the front axle. A worn range fork doesn't explain a failure to engage 4WD unless the question specifically says one actuator serves both functions — read carefully before answering a Tech A/B question built around this.

Planetary gearset — the rule that generates the high/low ratio: a simple set has a sun gear (center), a ring/internal gear (outer, teeth cut inward), and a planet carrier (pinions meshing with both). The set only produces a fixed ratio when exactly **one** member is held stationary — apply input to a second member, take output from the third. Hold nothing, and torque still passes through but there's no fixed ratio (effectively neutral). Lock any **two** members together and

the whole set turns as a unit: direct drive, 1:1. Holding the ring (sun in, carrier out) gives reduction; holding the carrier (ring in, sun out) gives overdrive and can reverse output direction. The *pattern* — which member is held, and what that produces — is the transferable fact; a specific class example's numeric ratio is not.

Engagement won't complete? Work the chain in order: driver/automatic request recognized → required speed/gear/brake/neutral conditions met → module has valid inputs, no inhibiting fault → command reaches the actuator under load → ground/return path carries current → motor/solenoid/vacuum/hydraulic mechanism actually moves → fork/cam/clutch completes mechanical engagement → position feedback agrees → calibration/initialization completes. A meter reading battery voltage at one motor terminal doesn't prove correct voltage *across* the motor under load — measure with the circuit operating.

Don't overlook tires: mismatched circumference (wrong size, uneven wear, a temporary spare) forces a persistent speed difference that can mimic a binding coupling or transfer case. Check placard size, all four installed sizes, inflation, and tread depth before condemning hardware.

Fluid: transfer-case lubricant is application-specific — “every transfer case uses ATF” is an unsafe assumption. The task list also covers shifting mechanisms, shafts/joints, wheel hubs, vents, and calibration/reset after service.

Quick Tech A/B method

Evaluate each technician's statement independently against the stated symptom before comparing them to each other. Ask: does this mechanism directly explain the described symptom, or does it explain a *different* function that only sounds related (range vs. mode, pinion depth vs. backlash, balance vs. runout vs. phasing vs. working angle)? The exam rewards keeping adjacent-but-different mechanisms separate more than it rewards memorizing part names.

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